

Utility-Specific Carbon Intensity of Electricity

Clean Fuel Standard program utility-specific electricity for 2025

Background

Ecology must annually publish the carbon intensity (CI) values for each electric utility in Washington and the statewide average grid electricity CI based on the mix of resources used to generate electricity using the most recent year's Fuel Mix Disclosure published by the Washington Department of Commerce (Commerce) (WAC 173-424-630(1)[1]). Ecology calculated the utility-specific and statewide grid average carbon intensity values for electricity using the Fuel Mix Disclosure for the electricity consumed in calendar year 2023 as reported by Commerce.

The carbon intensity values in this document are intended for use during quarterly and annual reporting in the Clean Fuel Standard for electricity consumed in calendar year 2025. The carbon intensity value of utility-specific electricity is used for reporting electricity used to fuel a vehicle in the specific utility service area in the Washington Fuel Reporting System according to Chapter 173-424 WAC.

Methodology

The 2025 utility-specific carbon intensity values of electricity were determined using Commerce's 2023 utility Fuel Mix Disclosure report, which contains the composition of resources used to generate electricity used in each utility service area. Commerce shared the 2023 annual report on December 19, 2024 after addressing the feedback from electric utilities, and will publish the final Fuel Mix Disclosure on its Fuel Mix Disclosure and Greenhouse Gas Content website[2]. The methodology is the same as the one used in 2024[3].

Carbon intensity accounts for the greenhouse gas emissions throughout the lifecycle of the electricity, from generation to the point of consumption. This document separates the carbon intensity of the upstream processes (before the electricity generation, including sourcing generation resources) from the downstream processes (from electricity generation through consumption). The calculation was performed using the WA Utility Carbon Intensity Calculator[4] developed during the 2022 rulemaking by the consulting firm Life Cycle Associates. The unit of carbon intensity is a gram CO_2 equivalent per unit of energy of the electricity (kilowatt hours, kWh).

Carbon intensity values of utility-specific electricity

The table below shows the carbon intensity values for electricity to be used in 2025 in each utility service area or charge electric vehicles and to produce hydrogen using electrolysis for fuel cell vehicles according to

[1] https://app.leg.wa.gov/wac/default.aspx?cite=173-424&full=true#173-424-630
[2] https://www.commerce.wa.gov/energy-policy/electricity-policy/fuel-mix-disclosure/
[3] https://apps.ecology.wa.gov/publications/SummaryPages/2414014.html
[4] https://ecology.wa.gov/Air-Climate/Reducing-Greenhouse-Gas-Emissions/Clean-Fuel-Standard/Requirements#CI

Chapter 173-424 WAC[5]. The 2023 statewide average grid electricity was 63.42 gCO2e/MJ, which is practically the same as 2022 (63.39 gCO2e/MJ). The detailed calculation is provided in the utility-specific calculator posted on the Clean Fuel Standard program document library website[6] under the section titled 'Electric Utility-Specific Carbon Intensity'.

Electric Utility Name	Pathway Codes	Upstream Emission, gCO2e/MJ	Production to Consumption phase emission, gCO2e/MJ	Utility Specific CI, gCO2e/MJ
Alder Mutual Light	ELCWAAML2025	3.42	13.44	16.86
Benton County PUD #1	ELCWABEN2025	3.54	14.99	18.53
Benton Rural Electric Assn	ELCWABRA2025	3.42	13.45	16.86
Big Bend Electric Coop	ELCWABBC2025	6.22	25.65	31.86
City of Blaine	ELCWABLN2025	3.42	13.45	16.86
Centralia City Light	ELCWACCL2025	4.32	17.51	21.83
Chelan County PUD #1	ELCWACHP2025	5.57	24.00	29.57
Cheney Light Department	ELCWACLD2025	5.20	21.20	26.40
Chewelah Electric Department	ELCWACED2025	3.42	13.45	16.86
Clallam County PUD #1	ELCWACCP2025	3.39	13.32	16.71
Clark County PUD #1	ELCWACLP2025	12.01	50.96	62.97
Clearwater Power (WA)	ELCWACWP2025	3.53	13.95	17.48
Coulee Dam, Town of	ELCWACOD2025	3.42	13.44	16.86
Cowlitz County PUD #1	ELCWACOP2025	3.91	14.52	18.43
Douglas County PUD #1	ELCWADCP2025	28.91	124.48	153.38
Eatonville Electric Department	ELCWAEED2025	3.42	13.44	16.86
Elmhurst Mutual Power & Light	ELCWAEML2025	3.42	13.45	16.86
Ellensburg Electric Division	ELCWAELD2025	3.42	13.45	16.86
Ferry County PUD #1	ELCWAFCP2025	3.42	13.45	16.86
Franklin County PUD #1	ELCWAFRP2025	4.86	19.78	24.64
Grays Harbor County PUD #1	ELCWAGHP2025	3.86	15.44	19.31
Inland Power & Light	ELCWAIPL2025	2.99	11.75	14.74
Klickitat County PUD #1	ELCWAKCP2025	6.41	26.62	33.04

[5] https://app.leg.wa.gov/WAC/default.aspx?cite=173-424&full=true
[6] https://www.ezview.wa.gov/site/alias__1962/37847/clean_fuel_standard_guidance_documents.aspx

Lakeview Light & Power	ELCWALAK2025	3.42	13.45	16.86
Lewis County PUD #1	ELCWALCP2025	5.20	21.25	26.46
McCleary Light & Power	ELCWAMCL2025	3.42	13.45	16.86
Milton Electric Division	ELCWAMED2025	3.42	13.44	16.86
Modern Electric Water Company	ELCWAMEW2025	3.42	13.45	16.86
Nespelem Valley Elec Coop	ELCWANCV2025	3.42	13.45	16.86
Northern Lights (WA)	ELCWANTL2025	3.75	14.90	18.65
Ohop Mutual Light	ELCWAOML2025	3.42	13.45	16.86
Okanogan County PUD #1	ELCWAOCP2025	13.64	57.96	71.60
Okanogan County Electric Coop	ELCWAOCC2025	3.80	15.12	18.93
Pacific County PUD #2	ELCWAPCP2025	6.01	24.76	30.78
Parkland Light & Water	ELCWAPLW2025	3.42	13.45	16.86
Grant County PUD #2	ELCWAGCP2025	24.58	105.85	130.43
Pend Oreille County PUD #1	ELCWAPOP2025	22.66	97.27	119.94
Peninsula Light	ELCWAPEL2025	3.42	13.45	16.86
Asotin County PUD #1	ELCWAACP2025	3.42	13.45	16.87
Port Angeles Light Operations	ELCWAPAL2025	3.42	13.45	16.86
Wahkiakum County PUD #1	ELCWAWCP2025	3.42	13.45	16.86
Mason County PUD #3	ELCWAMCP2025	2.95	11.59	14.54
Puget Sound Energy	ELCWAPSE2025	11.79	90.29	102.08
Richland Energy Services	ELCWARES2025	4.78	19.42	24.20
Ruston, Town of	ELCWARUS2025	1.93	7.57	9.49
Seattle City Light	ELCWASCL2025	2.41	10.16	12.58
Skamania County PUD #1	ELCWASCP2025	3.42	13.45	16.86
Snohomish County PUD #1	ELCWASNO2025	3.07	11.74	14.81
Steilacoom Electric Utility	ELCWASEU2025	3.42	13.45	16.86
Sumas, City of	ELCWASUM2025	3.42	13.45	16.86
Tacoma Power	ELCWATPO2025	1.93	7.57	9.49
Tanner Electric Coop	ELCWATEC2025	3.42	13.45	16.86
Vera Water & Power	ELCWAVER2025	4.42	17.83	22.25
Avista (WA)	ELCWAAVI2025	14.89	93.03	107.92
Mason County PUD #1	ELCWAMAP2025	3.17	12.48	15.66

Whatcom County PUD #1	ELCWAWHP2025	3.42	13.45	16.86
Jefferson County PUD #1	ELCWAJCP2025	3.42	13.45	16.86
Port of Seattle	ELCWAPOS2025	3.42	13.45	16.86
Yakama Power	ELCWAYKM2025	3.42	13.45	16.86
Pacific Power (WA)	ELCWAPAP2025	17.03	151.72	168.75
Kalispel Tribal Utility	ELCWAKTU2025	3.42	13.45	16.86
Orcas Power & Light Coop	ELCWAOPC2025	3.57	14.13	17.71
Energy Northwest	ELCWAENN2025	3.42	13.44	16.86
Consolidated Irrigation District #19	ELCWACID2025	3.41	13.42	16.83
Fairchild Airforce Base	ELCWAFAB2025	3.42	13.45	16.86
Grant County PUD #2 (Rate Schedule 13)	ELCWAGRP2025	0.00	0.00	0.00
Puget Sound Energy Green Direct Program	ELCWAPSR2025	0.00	0.00	0.00
Douglas County PUD #1 (Schedule 4 – Renewable)	ELCWADCR2025	0.00	0.00	0.00
WA State Average Electric Grid	**WAELEC001-25**	**9.27**	**54.15**	**63.42**

Email comments to:

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